

# Software Requirements Specification (SRS) Language Translation Web Application (NLP-Based)

## 1. Introduction

**Purpose:** This document describes requirements for a web-based language translation system using NLP techniques such as Seq2Seq and Transformer models.

**Scope:** The application translates text between multiple languages in real time via a web interface.

**Definitions:** NLP – Natural Language Processing; Seq2Seq – Sequence to Sequence; UI – User Interface; API – Application Programming Interface.

## 2. Overall Description

**Product Perspective:** Standalone web app consisting of frontend, backend, and translation model.

**Product Functions:** Accept input text, translate to target language, display output, handle errors.

**User Classes:** General users, students, professionals, researchers.

**Operating Environment:** Web browser, Python backend (Flask/Django), PyTorch/TensorFlow.

## 3. System Features

**Text Translation:** Converts source language text into target language using Seq2Seq or Transformer.

**Language Selection:** Users choose source and target languages.

**Automatic Detection (Optional):** System detects source language automatically.

**Error Handling:** Handles empty input, unsupported languages, and server issues.

**Translation History (Optional):** Stores recent translations.

## 4. External Interface Requirements

**User Interface:** Text area input, language dropdowns, translate button, output display.

**Hardware Interface:** Standard computer or mobile device.

**Software Interface:** NLP frameworks and web frameworks.

**Communication Interface:** HTTP/HTTPS via browser.

## 5. Non-Functional Requirements

**Performance:** Response within ~3 seconds for short text.

**Reliability:** High availability and accuracy.

**Security:** Input validation and HTTPS.

**Usability:** Simple, user-friendly interface.

Maintainability: Modular design and easy updates.

Scalability: Support more languages and cloud deployment.

## 6. System Architecture

Frontend → Backend Server → NLP Model → Translated Output

## 7. Future Enhancements

Speech input/output, offline mode, mobile app, document translation.

## 8. Acceptance Criteria

System is accepted if translation works correctly, UI responds, errors handled, and performance targets met.